# FARJANA FERDAUSI

**AI/ML Engineer | Machine Learning, Deep Learning, NLP, LLMs, RAG**

Dinajpur, Bangladesh | +880 1675-466770 | farjana.rafi1983@gmail.com

LinkedIn: https://www.linkedin.com/in/farjana-ferdausi/ | GitHub: https://github.com/farjanaferdausi-cs50ai

Medium: https://medium.com/@farjana.rafi1983 | Portfolio: https://farjanaferdausi-cs50ai.github.io/Farjana-Ferdausi-Resume/

## SUMMARY

AI/ML Engineer building grounded, cloud-deployed LLM systems: RAG assistants, multi-agent workflows, data agents and computer-vision services using Python, Gemini, LangChain, LangGraph, FastAPI, Docker and Google Cloud. Google Cloud Gen AI Academy APAC (Cohort 3) Fellow with 26 skill badges and 6 AI applications deployed on Cloud Run. Former HR professional with 14+ years of experience, bringing disciplined documentation, interview coordination, reporting and stakeholder communication to AI/ML engineering. Seeking AI/ML Engineer roles owning systems from prototype to deployment.

## TECHNICAL SKILLS

**Languages & Data:** Python (pandas, NumPy, Matplotlib, Pydantic), SQL, REST APIs, Data pipelines / ETL

**Machine Learning:** scikit-learn, supervised and unsupervised learning, feature engineering, cross-validation, model evaluation

**Deep Learning, NLP & CV:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, BERT fine-tuning, embeddings, OpenCV, YOLOv8, YOLOv11, object tracking (Deep SORT)

**LLMs & Generative AI:** RAG, hybrid retrieval (BM25 + embeddings), ChromaDB, prompt engineering, structured outputs, tool calling, MCP, multi-agent systems, Gemini, LangChain, LangGraph, CrewAI, Google ADK, LangSmith, n8n

**MLOps & Cloud:** Docker, FastAPI, Git, CI/CD, Google Cloud (Vertex AI, Cloud Run, BigQuery, IAM, Secret Manager), Firebase (Auth, Firestore), Google Maps Platform, Google Earth Engine, Render, Vercel

## EXPERIENCE

**AI/ML Engineer — Freelance** | TR Enterprise (Start-up), Technology Division, Remote **1 Sep 2026 – Present**

- Design, build and deploy cloud-hosted RAG chatbots, computer-vision services and data-analysis agents using Gemini, Google ADK, LangChain, LangGraph, BigQuery, FastAPI and Docker, owning the ML lifecycle from prototype to deployment.
- Containerize FastAPI services with Docker and ship them to Google Cloud Run.
- Enforce safeguards in AI workflows: least-privilege IAM, read-only SQL with dry-run validation, and human approval before any write action.

**AI/ML Engineering Fellow** | Google Cloud Gen AI Academy APAC Edition (Cohort 3) **1 Aug 2026 – 30 Sep 2026**

- Completed all three tracks plus Meet the Builders, the Ideathon and the AI Builder Cup 2026 Hackathon.
- Delivered 6 AI applications on Google Cloud Run: RAG Barista agent, BigQuery MCP data agent, AI Coffee Manager, Flood Risk Assistant Bangladesh, AI/ML Learning Journal and OmniClean AI.

**Artificial Intelligence Intern** | CodeAlpha, India (Remote) **20 Jul 2026 – 20 Aug 2026**

- Delivered 4 end-to-end AI projects: a 20-language voice translator, a TF-IDF/cosine-similarity FAQ chatbot, an LSTM Bach-chorale generator, and YOLOv8 + Deep SORT object tracking.
- Shipped each as a documented, runnable GitHub repository; received a Letter of Recommendation.

**Career Break & Continuous Learning** | Self-directed, Bangladesh **4 Oct 2020 – Feb 2026**

*Planned, self-directed transition from HR to AI/ML engineering through structured, certified learning.*

- **Upskilled in HR:** two Alison diplomas (Modern HRM, 2022; Human Resources, 2023) and three certificates (International & Strategic HRM; HR Policies & Organization Structure; HR in Food Service & Hospitality; 2022).
- **Built GenAI foundations:** AI Content Creation Bootcamp (Batch 82) and FB & YT Monetization (Batch 83), SR Dream IT, May – Jul 2025; Generative AI Mastermind (Outskill, 2025); Introduction to AI (Alison, 2026).
- **Communication: Continuous English development, Keith Speaking Academy (2 Oct 2023 – Present)**, covering IELTS Speaking GOLD Interactive, Fluent Grammar, Vocabulary Vault and Idioms courses.

**HR Coordinator** | Radisson Blu Dhaka Water Garden, Dhaka, Bangladesh **15 Mar 2006 – 3 Oct 2020**

- Screened resumes and candidates, scheduled and conducted interviews, and coordinated joining formalities, induction and training for new employees.
- Administered leave and daily time and attendance through the HRIS, tracking late arrivals, early departures and absenteeism.
- Maintained employee files and HR records for 400+ employees with 100% documentation compliance; built Excel headcount and attrition reports.
- Coordinated employee engagement and fundraising events; recognized with the Women's Loyalty Award (2018) and multiple Radisson Diamond, Platinum and Gold Awards.

## PROJECTS

**OmniClean AI: Waste Intelligence Platform** | (Sustainable and Social Impact) **1 Oct 2026 – 18 Oct 2026**

*AI Builder Cup 2026, Google Cloud & Hack2skill (Hackathon)*

- Built an AI waste-intelligence platform that identifies items in a photo with Gemini, gives sorting guidance in English and Bengali, blocks recycling advice for hazardous items (batteries, chemicals, sharps, pressurized cans) and flags scans below 0.6 confidence for retake.

- Added an impact dashboard (recycling rate, estimated $CO_2e$ avoided using EPA WARM v15 factors, 7-day trend) and a garment-factory monitor with a waste-per-1,000-pieces KPI and anomaly detection (> mean + 2σ) across 5 departments.
- Streamed analytics into 4 date-partitioned BigQuery tables using parameterized SQL and Application Default Credentials (no keys in source); kept the Gemini key server-side with a 4-model fallback ladder; deployed on Google Cloud Run.

**Tech:** Gemini, Vertex AI, React, TypeScript, Express, Firebase, Firestore, BigQuery, Cloud Run
**GitHub:** https://github.com/farjanaferdausi-cs50ai/OmniClean-AI
**Live:** https://omniclean-ai-465358201476.asia-southeast1.run.app/

**ESBC-AI: Easy Service for Bangladeshi Citizens** | Ostad Institute, Capstone Project (Module 27) **19 Oct 2026 – 28 Oct 2026**

- Built a bilingual (Bangla and English) RAG assistant giving citation-grounded answers on passport, NID, birth and death registration, fees, tax and social allowances, using only official .gov.bd pages stored with source URL and verification date.
- Curated a 29-document knowledge base across 7 service areas; designed hybrid retrieval (Gemini embeddings + BM25, reciprocal rank fusion) with ChromaDB and a LangGraph agent with a confidence gate (0.45 threshold) plus a NO_ANSWER refusal protocol.
- Evaluated on a fixed 29-question bilingual set (keyword-based): 100% top-5 retrieval hit rate, 100% answer accuracy and 100% correct refusal on out-of-scope questions; offline pytest suite covers retrieval, API contract and refusal path.
- Served via FastAPI with Gemini model fallback and response caching; containerized with Docker and deployed on Render (API) and Vercel (frontend).

**Tech:** Python, FastAPI, LangGraph, RAG, Gemini embeddings, BM25, ChromaDB, Docker, Render, Vercel
**GitHub:** https://github.com/farjanaferdausi-cs50ai/ESBC-AI
**Live:** https://esbc-ai.vercel.app

**AI/ML Learning & Career Journal** | Google Cloud Gen AI Academy APAC Edition (Cohort 3), Ideathon **22 Aug 2026 – 6 Sep 2026**

- Developed an authenticated web app with multi-turn Gemini coaching, 7-part structured session summaries and a searchable timeline across four learning programs; added weak-skill detection, career-gap analysis and trend analysis over past sessions.
- Integrated Google Maps Platform (Places search, geolocation) for study-location tagging; secured data with Firebase Auth, owner-bound Firestore rules and Secret Manager (Gemini and Maps keys); deployed on Cloud Run.

**Tech:** Gemini API, React, Node.js, Express, Firebase Auth, Firestore, Google Maps Platform, Secret Manager, Cloud Run
**GitHub:** https://github.com/farjanaferdausi-cs50ai/AI-ML-Learning-Career-Journal-CloudRun
**Live:** https://ai-ml-learning-career-journal-1.ai.studio

## EDUCATION & TRAINING

**AI & ML Engineer Bootcamp, Batch 6** | Ostad, Bangladesh **18 Feb 2026 – Present**

**AI Engineering & Data Science Bootcamp 4.0** | Codebasics, India **22 Jul 2026 – Present**

**M.A., English Language & Literature** | Bangladesh University (CGPA 3.52 / 4.00) **2013 – 2014**

**B.A., English Language & Literature** | International Business Administration and Information System University (CGPA 3.25 / 4.00) **2008 – 2012**

## CERTIFICATIONS

- Google Cloud Gen AI Academy, APAC Edition (Cohort 3), Fellow — Hack2skill & Google Cloud, Sep 2026 (ID 2026H2S09GCGENAIAPACC3-P00697)
- Google Cloud Skills Boost — 26 skill badges (2026), including Gemini for App Developers, MLOps for Generative AI, Model Armor, Responsible AI, Introduction to AI Agents and Attention Mechanism
- CodeAlpha Artificial Intelligence Internship, with Letter of Recommendation — Aug 2026 (ID CA/DF1/205038)
- Generative AI Mastermind — Outskill, May 2025 & Jun 2026 (completed twice) | Advanced Prompt Engineering — Simplilearn, 2026
- Python for Data Science & ML — UniAthena, 2026 | Introduction to Artificial Intelligence — Alison, Feb 2026 (ID 25255027)
- Human Resources — Alison: Diploma in HR (Nov 2023), Diploma in Modern HRM (Jun 2022) and three short HR courses (2022)

## VOLUNTEERING & LEADERSHIP

- **SOS Children's Village** | HR Team Representative, Radisson Blu | 2016 – 2020: visited about once every two months to donate food, clothes, soap and funds, support the children's education and spend time with them.
- **Crisis Management Team Member** | Radisson Blu | Aug 2015 – Oct 2020: supported emergency preparedness; CPR and First Aid certified by Bangladesh Red Crescent Society (2009, refresher 2011); Fire and Safety trained with annual refreshers (2007–2019).
- **Social Committee Member** | Radisson Blu | Dec 2015 – Oct 2020: organized engagement activities and fundraising for relief.
- **Employee Child Care Center Supervisor** | Radisson Blu | Oct 2009 – Oct 2020: managed budget, hygiene inspections and daily activities for children aged 0–5.